

Securitize Public Stock On/Off Ramp Audit Summary

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Version 2.0

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Securitize on Securitize Public Stock On/Off Ramp. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated December 24, 2025, was delivered to Securitize, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 8 days, the Cyfrin team conducted an audit on the Securitize Public Stock On/Off Ramp code provided by Securitize. The findings consist of 4 Medium & 14 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Securitize Public Stock On/Off Ramp
Repository	bc-on-off-ramp-sc
Commit	a2935dbffb53...
Fix Commit	f0fff37f39c2...
Repository 2	bc-nav-provider-sc
Commit	c31fc8d3621b...
Fix Commit	d05482b2d42a...
Audit Timeline	Dec 1st - Dec 10th, 2025
Methods	Manual Review, Invariant Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	4	4	0
Low Risk	14	11	3
Informational	10	4	6
Gas Optimizations	13	7	6
Total	41	26	15

6 Audit Scope

The audit scope was limited to:

```
// new and modified files
bc-on-off-ramp-sc/contracts/on-ramp/PublicStockOnRamp.sol
bc-on-off-ramp-sc/contracts/on-ramp/BaseOnRamp.sol
bc-on-off-ramp-sc/contracts/on-ramp/SecuritizeOnRamp.sol
bc-on-off-ramp-sc/contracts/off-ramp/BaseOffRamp.sol
bc-on-off-ramp-sc/contracts/off-ramp/PublicStockOffRamp.sol
bc-on-off-ramp-sc/contracts/off-ramp/SecuritizeOffRamp.sol
bc-nav-provider-sc/contracts/nav/SecuritizeAmmNavProvider.sol

// existing files
bc-on-off-ramp-sc/contracts/common/BaseContract.sol
bc-on-off-ramp-sc/contracts/fee/MbpsFeeManager.sol
bc-on-off-ramp-sc/contracts/on-ramp/provider/AllowanceAssetProvider.sol
bc-on-off-ramp-sc/contracts/on-ramp/provider/MintingAssetProvider.sol
bc-on-off-ramp-sc/contracts/off-ramp/provider/AllowanceLiquidityProvider.sol
bc-on-off-ramp-sc/contracts/off-ramp/provider/CollateralLiquidityProvider.sol
bc-on-off-ramp-sc/contracts/off-ramp/CountryValidator.sol
bc-on-off-ramp-sc/contracts/off-ramp/RedemptionManager.sol
bc-on-off-ramp-sc/contracts/off-ramp/RedemptionValidator.sol
bc-on-off-ramp-sc/contracts/off-ramp/TokenCalculator.sol
bc-nav-provider-sc/contracts/nav/RedStoneNavProvider.sol
bc-nav-provider-sc/contracts/nav/SecuritizeInternalNavProvider.sol
bc-nav-provider-sc/contracts/utils/Helper.sol
bc-nav-provider-sc/contracts/utils/BaseContract.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Securitize is developing new functionality to enable blockchain-based trading of tokenized public stocks (TSLA, AAPL) through DS Token representations. This audit covers two repositories that together form the core infrastructure for pricing and investor onboarding/offboarding: the NAV Provider and the On/Off Ramp components.

8.1 NAV Provider

The new addition to NAV Provider is `SecuritizeAmmNavProvider`, a virtual Automated Market Maker (AMM) based on the Uniswap V2 constant product formula ($x \times y = k$). Unlike traditional AMMs, this implementation uses virtual reserves that track "inventory pressure" without holding actual tokens. The system combines an external anchor price (from oracles representing real market prices) with AMM-derived price deviation to create execution prices that reflect both real-world value and supply/demand dynamics.

The virtual AMM serves as Securitize's inventory management mechanism. When investors heavily buy tokenized securities, the deviation multiplier increases, creating a premium above the oracle price that incentivizes selling. Conversely, heavy selling creates discounts that attract buyers. A configurable scale factor dampens these de-

viations to prevent excessive price impact. The system supports two market states (OPEN and CLOSED), with baseline resets occurring on market transitions to re-center the AMM around the current anchor price.

8.2 On/Off Ramp

The On/Off Ramp repository provides the entry and exit infrastructure for investors to acquire and redeem tokenized securities. The on-ramp contracts (`BaseOnRamp`, `PublicStockOnRamp`, `SecuritizeOnRamp`) handle the conversion of liquidity tokens (typically stablecoins such as USDC) into DS Tokens representing tokenized securities. The off-ramp contracts (`BaseOffRamp`, `PublicStockOffRamp`, `SecuritizeOffRamp`) handle the reverse process, allowing investors to redeem their DS Tokens for liquidity tokens.

Compliance features include investor registry validation through Securitize's DS Protocol services and country-based redemption restrictions. The contracts support configurable liquidity providers (`AllowanceLiquidityProvider`, `CollateralLiquidityProvider`, `MintingAssetProvider`) and fee management, with both single-step and two-step transfer modes.

9 Attestation

Cyfrin attests that it performed a security review of the Securitize Public Stock On/Off Ramp code provided by Securitize.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-12-24-cyfrin-securitize-public-stock-ramp-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Securitize has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=26d7273cb3e7e9a0e10bac3a4761efd44b3d17c206c1c11062c6f194ea114524
F=2025-12-24-cyfrin-securitize-public-stock-ramp-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.